

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
SECJ3104 APPLICATION DEVELOPMENT

SECTION 02

SEMESTER 1 2025/2026

PROJECT PROPOSAL

PROJECT TITLE: ThriftIn UTM

SUBMISSION DATE: 28th OCTOBER 2025

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1.0 EXECUTIVE SUMMARY

In this project, ThriftIn UTM is proposed as a comprehensive mobile application designed to establish a secure, authenticated and highly efficient digital marketplace specifically for students of Universiti Teknologi Malaysia (UTM). The platform's primary function is to facilitate the buying and selling of preloved items such as clothes, books, electronics and room essentials easily, safely and efficiently within the UTM campus community. This dedicated solution directly addresses the significant limitations encountered when using existing informal communication channels, such as general university Telegram and WhatsApp groups, where commercial activities are often treated as off-topic and subject to deletion or member removal by group administrators. By centralizing the marketplace into a specialized digital environment, the platform actively promotes a sustainable reuse culture and strengthens student entrepreneurship by offering a dedicated channel for small-scale commerce. Furthermore, ThriftIn UTM simplifies transactions through digital integration and is built around key features, including mandatory user authentication and authorization, a structured transaction flow, advanced data analytics, and predictive AI integration. ThriftIn UTM is positioned to become the official, reliable, and primary platform for campus commerce, fostering community trust, and enhancing affordability for the entire student body.

2.0 PROJECT DOMAIN AND COMMUNITY BACKGROUND

The Universiti Teknologi Malaysia (UTM) student community currently operates without a centralized, authorized platform dedicated to second-hand and preloved items selling and buying. The current method involves students using informal digital channels, primarily residential college messaging groups, which are fundamentally designed for official announcements and administrative communication. This reliance on unsuitable platforms creates several systemic issues. Firstly, it leads to inconsistent policy enforcement as selling or buying posts are frequently deleted by group administrators, creating issues and frustrating users who simply need it. Secondly, there is a critical lack of trust and security inherent in these open groups, making it difficult for users to verify a seller's genuine student status or to ensure the quality and authenticity of the item being sold. Finally, the inefficiency of the item discovery is a major problem, as product listings are rapidly overwhelmed and lost beneath new administrative messages and announcements, rendering effective searching or browsing virtually impossible. The entire initiative to create this structured, secure marketplace is proposed to be managed and overseen by UTM Entrepreneurship Club (*Kelab Keusahawanan UTM*), which is committed to nurturing the entrepreneurial spirit and facilitating student commerce activities on campus. ThriftIn UTM operates within the high-impact domain of campus commerce and sustainability, offering a structured, professional and scalable solution aimed at significantly enhancing student welfare and actively supporting the circular economy within the UTM ecosystem.

3.0 PROBLEM ANALYSIS

To gather information about the problems faced by the community, our team has conducted distributed questionnaires.

3.1 QUESTIONNAIRES

We distributed our questions through social media platforms such as Facebook, Instagram and several WhatsApp groups to reach UTM students from different levels. Eventually, we managed to gather 31 responses.

3.1.1 DEMOGRAPHIC INFORMATION

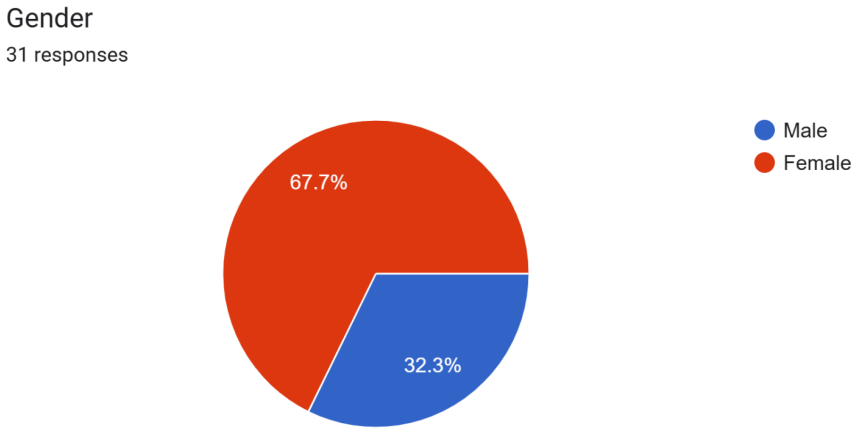


Figure 3.1: Gender distribution of respondents

Based on Figure 3.1, a total of 31 respondents participated in the survey. Among them, 67.7% were female, while 32.3% were male. This indicates that the majority of participants were female students. The gender distribution suggests that female students were more responsive to the survey or may show a higher level of interest in buying and selling preloved items compared to their male counterparts.

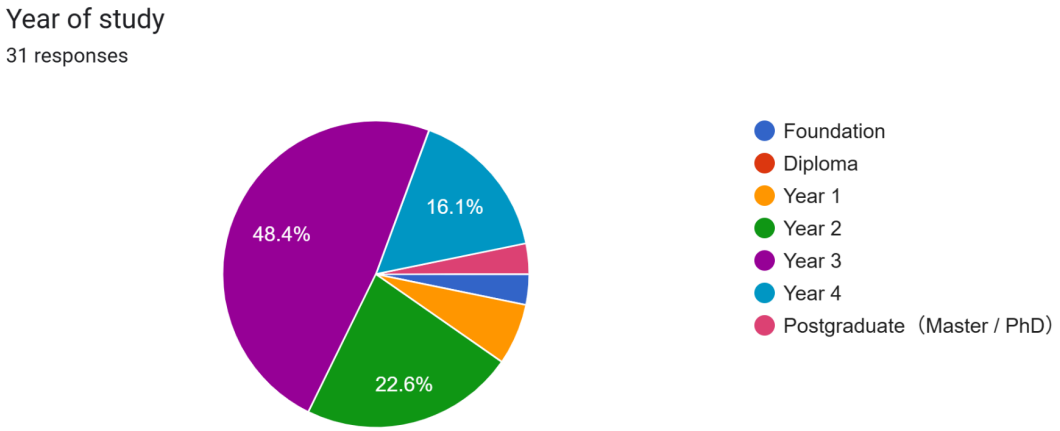


Figure 3.2: Year of study of distribution

As shown in Figure 3.2, most of the respondents are Year 3 students, making up 48.4% of the total. This is followed by Year 2 students with 22.6% and Year 4 students with 16.1%. Only a few respondents are from Year 1, Diploma, Foundation and Postgraduate levels, showing that most participants are from higher years and have more experience studying at the university.

3.1.2 BUYING AND SELLING EXPERIENCE

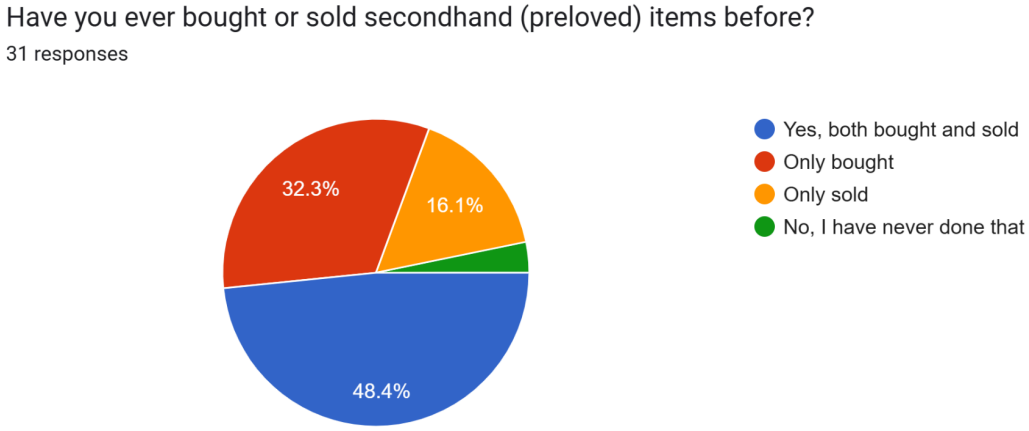


Figure 3.3: Respondent's experience in buying or selling preloved items

Figure 3.3 shows that nearly half of the university student respondents (48.4%) have both bought and sold secondhand items, indicating active engagement in the preloved market. Additionally, 32.3% have only purchased such items while 16.1% have only sold them. Only 3.2% have never participated in secondhand transactions. These results suggest that preloved item exchange is a

common practice among students. The high level of participation reflects a growing openness to circular consumption habits within the student community.

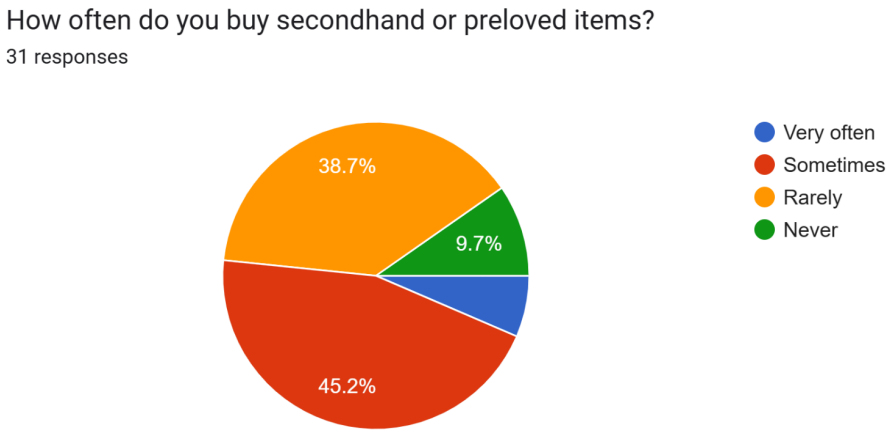


Figure 3.4: Frequency of buying preloved items

Figure 3.4 illustrates the frequency with which university students purchase secondhand or preloved items. The majority (45.2%) reported buying such items sometimes, followed by 38.7% who do so rarely. A smaller group (9.7%) indicated they buy very often while only 6.5% stated they never engage in secondhand shopping. These results suggest that most students are at least occasional participants in the preloved market, reflecting a growing awareness of sustainable consumption and budget-friendly alternatives among the student population.

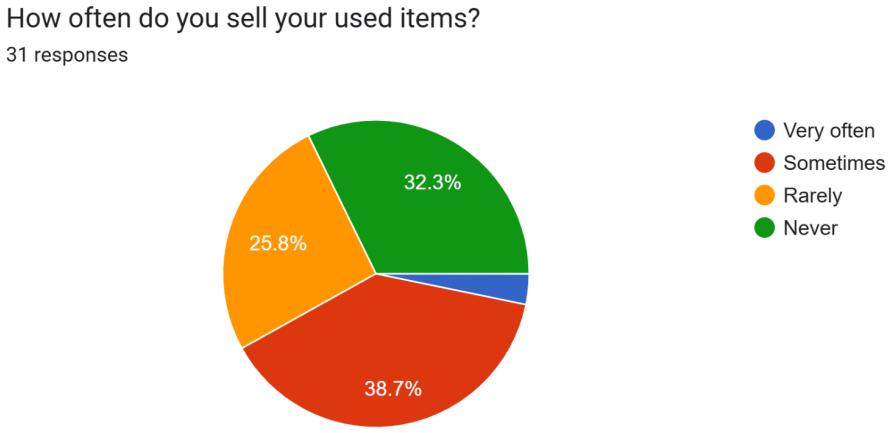


Figure 3.5: Frequency of selling used items

Figure 3.5 reveals that a significant portion of university students (38.7%) have never sold their used items, 25.8% do so rarely and 32.3% sometimes. Only a small minority (3.2%) reported selling very often. These findings suggest that while buying secondhand items is relatively common among students, selling used items is less frequent. The presence of occasional sellers indicates potential for growth in student-driven resale culture, especially with the rise of digital platforms and sustainability-focused initiatives.

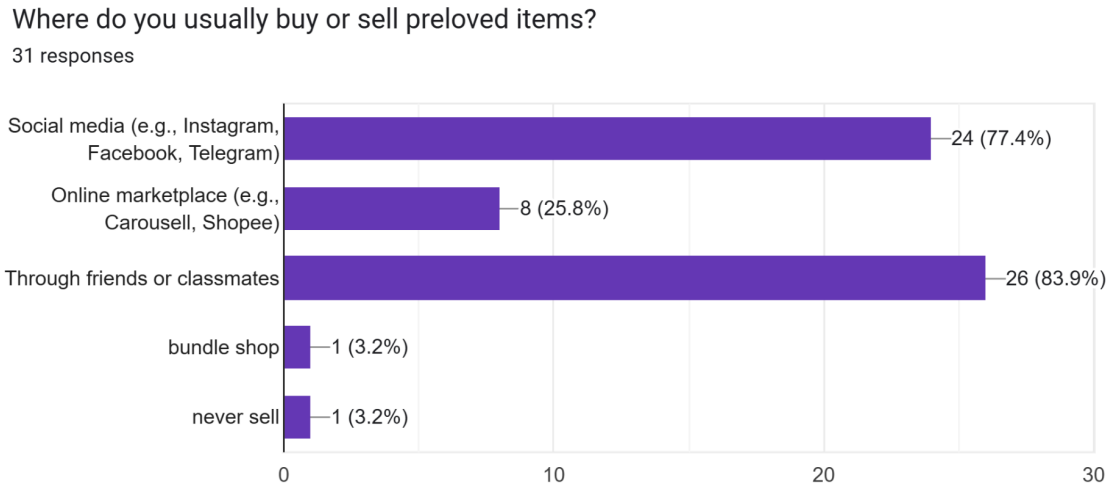


Figure 3.6: Common Platforms for buying or selling preloved items

Figure 3.6 highlights the most common platforms university students use to buy or sell preloved items. The majority of respondents (83.9%) engage in these transactions through friends or classmates, followed closely by social media platforms such as Instagram, Facebook, and Telegram (77.4%). Online marketplaces like Carousell and Shopee are used by 25.8% of students while only one respondent (3.2%) reported using bundle shops. Another 3.2% indicated they never sell preloved items. These findings suggest that informal, peer-to-peer channels are the preferred method among students, likely due to convenience, trust and ease of communication. This highlights a clear opportunity for a dedicated university-based platform that combines the trust and convenience of peer interaction with the structure and accessibility of digital tools.

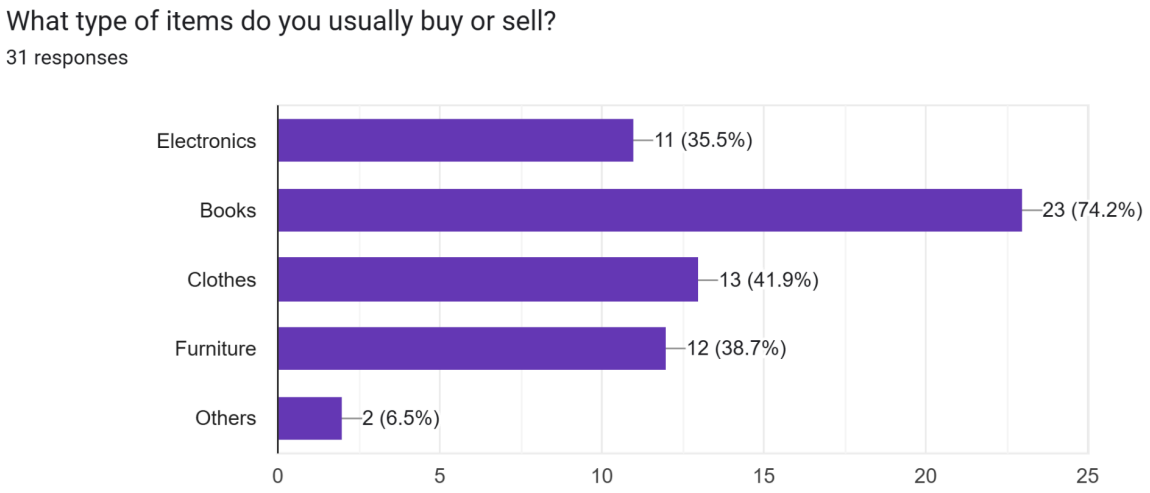


Figure 3.7: Types of items bought or sold by respondents

Figure 3.7 identifies the most commonly bought or sold items among university students with books leading at 74.2%, followed by clothes (41.9%), furniture (38.7%) and electronics (35.5%). Only 6.5% of respondents selected others, indicating that most secondhand activity revolves around academic and lifestyle essentials. These insights suggest that features should cater specifically to book exchanges, clothing resale, and dorm-related items like furniture and electronics. Prioritizing these categories can enhance platform relevance and engagement.

3.1.3 BUYING AND SELLING BEHAVIOR

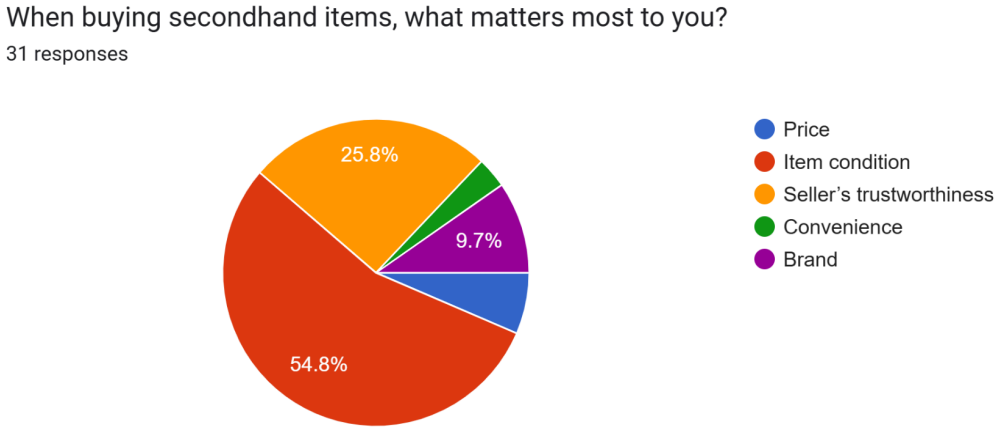


Figure 3.8: Important factors considered when buying preloved items

Figure 3.8 reveals that seller's trustworthiness is the most important factor for university students when purchasing secondhand items with 54.8% of respondents prioritizing it. This is followed by item condition (25.8%) while brand (9.7%), price (6.5%) and convenience (3.2%) are considered less critical. These insights highlight the importance of building trust and transparency between users on the platform.

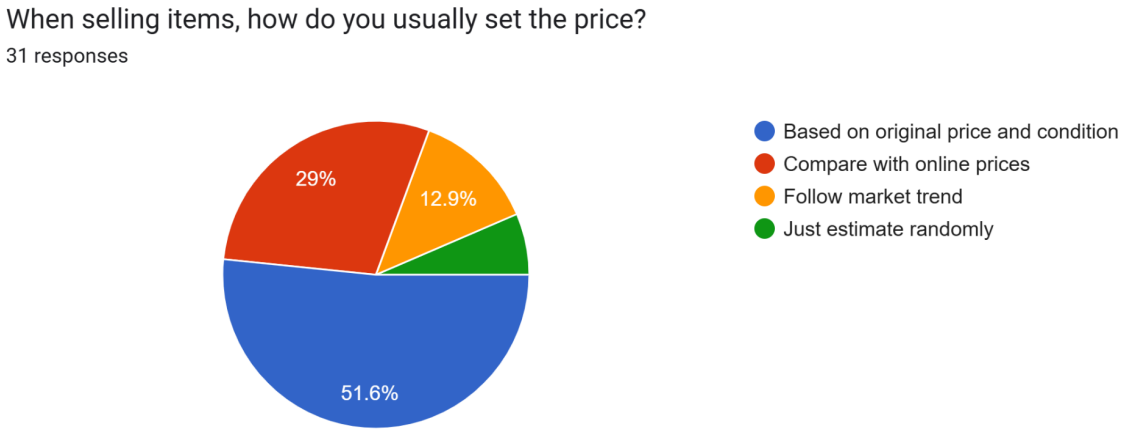


Figure 3.9: Methods used to determine selling price

Figure 3.9 shows that most university students (51.6%) set secondhand prices based on the original price and item condition while 29% compared with online listings. Smaller groups follow market trends (12.9%) or estimate randomly (6.5%). These findings suggest that students prefer practical, reference-based pricing methods.

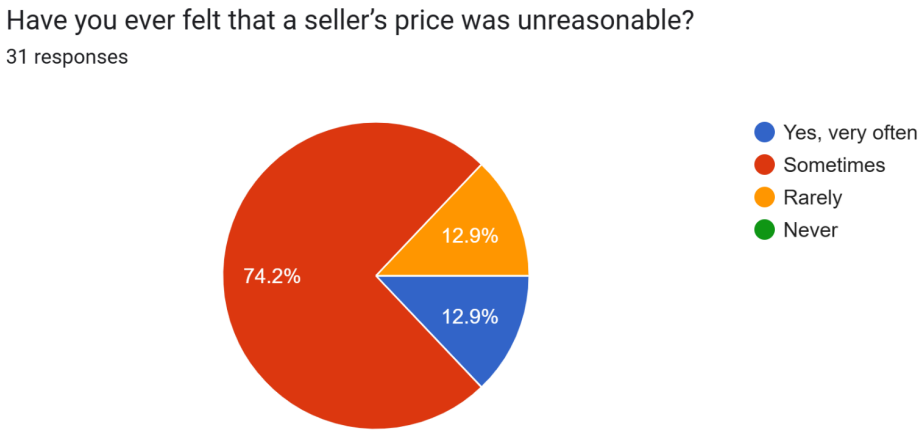
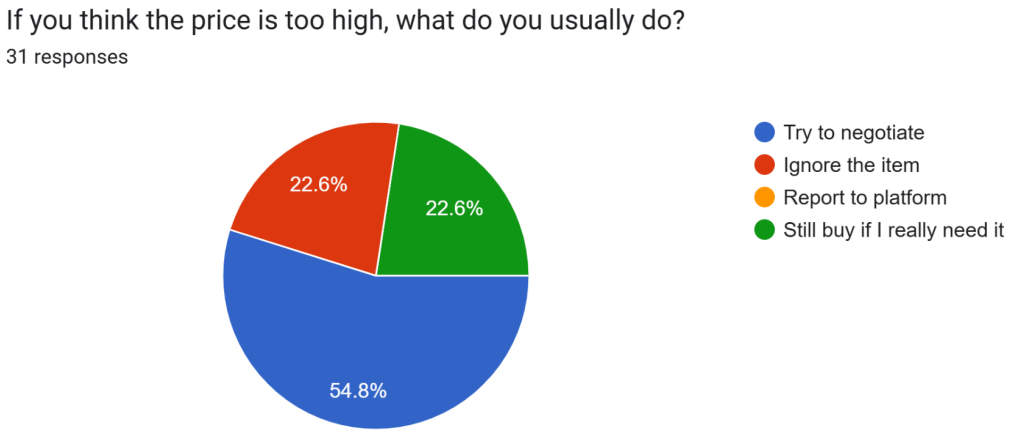


Figure 3.10: Respondents' perception of unreasonable seller pricing

Figure 3.10 shows that a majority of university students (74.2%) sometimes feel that seller pricing is unreasonable while 12.9% experience this very often and another 12.9% rarely.

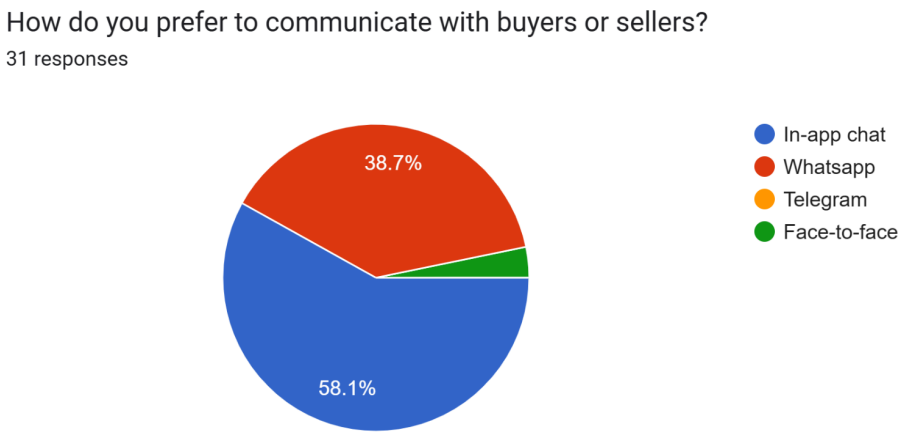
Notably, none of the respondents selected never, indicating that pricing concerns are a common issue in secondhand transactions. This highlights the need for clearer pricing standards and transparency features within the platform.



3.11: Actions taken when seller's price is too high

Figure 3.11 shows that when student respondents perceive a seller's price as too high, the majority (54.8%) choose to negotiate. The remaining respondents either ignore the item (22.6%) or still buy it if necessary (22.6%). Notably, no respondents selected a report to platform, indicating that formal complaint mechanisms are rarely used or possibly not expected in peer-to-peer secondhand exchanges. This suggests that user-friendly communication tools are essential to help reduce pricing friction and improve transaction outcomes.

3.1.4 COMMUNICATION AND PAYMENT PREFERENCES



3.12: Preferred communication methods in preloved transactions

Figure 3.12 shows that the majority of university students (58.1%) prefer to communicate via in-app chat when buying or selling preloved items, followed by WhatsApp (38.7%). Telegram and face-to-face communication were less preferred with lower response rates. These findings suggest that students favor integrated messaging over external apps or offline methods.Therefore, the platform should prioritize a built-in chat feature that supports real-time, secure communication between buyers and sellers.

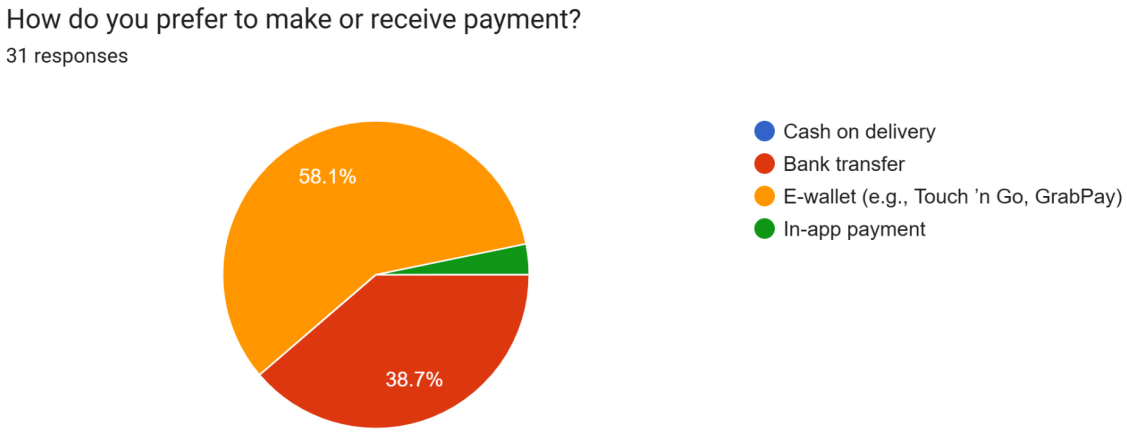


Figure 3.13: Preferred payment method for preloved transactions

Figure 3.13 shows that e-wallets are the most preferred payment method among university students (58.1%), followed by bank transfer (38.7%). Cash on delivery received no responses, and in-app payment was the least selected option. These findings suggest a strong preference for digital payment solutions, likely due to their convenience, speed and widespread use among students. For platform development, this highlights the importance of integrating secure e-wallet and bank transfer options.

3.1.5 CHALLENGES AND SATISFACTION

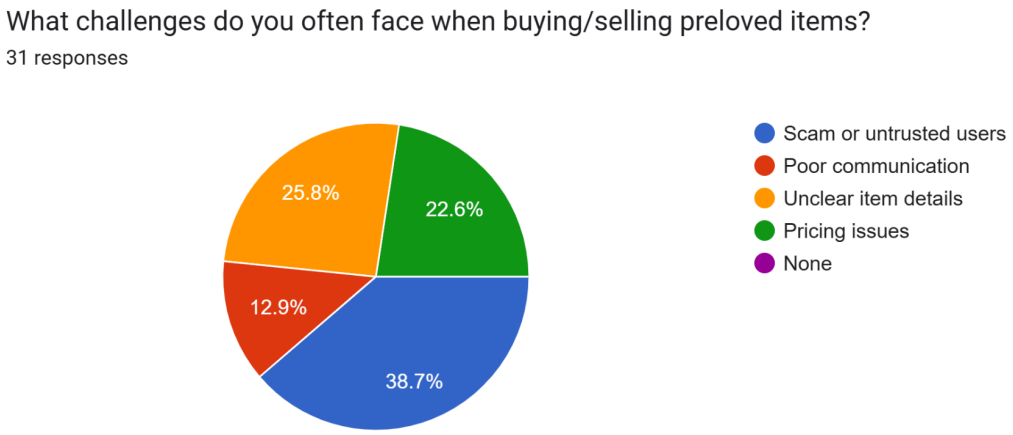


Figure 3.14: Common challenges faced in preloved transactions

Figure 3.14 shows that the most common challenge faced by university students in preloved transactions is dealing with scams or untrusted users (38.7%), followed by unclear item details (25.8%), pricing issues (22.6%) and poor communication (12.9%). No respondents selected none, indicating that most students encounter at least one issue when buying or selling secondhand items. These findings underscore the need for platform features that enhance trust and clarity such as verified student accounts to improve user experience and reduce transactional risks.

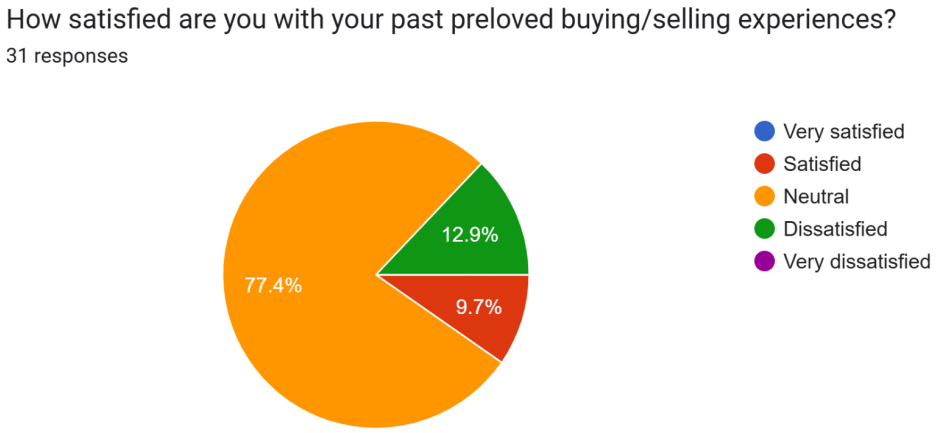


Figure 3.15: Satisfaction level with past preloved transactions

Figure 3.15 shows that the majority of university students (77.4%) reported a neutral level of satisfaction with their past preloved buying or selling experiences. Only 12.9% felt satisfied while 9.7% were dissatisfied. No respondents selected very satisfied or very dissatisfied, indicating that most experiences were neither strongly positive nor negative. This suggests that while secondhand transactions are generally acceptable. The predominance of neutral satisfaction suggests that enhancements in trust enhancement, communication tools and listing clarity could significantly improve user experience

3.1.6 TRUST AND PLATFORM PERCEPTION

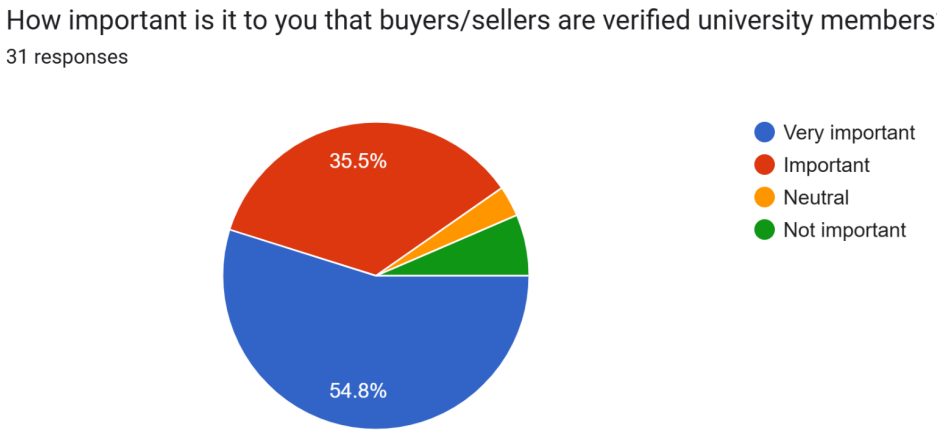


Figure 3.16: Importance of buyer/seller verification among university members

Figure 3.16 shows that the majority of university students (54.8%) consider it very important that buyers and sellers are verified university members while 35.5% view it as important. Only a small portion selected neutral or not important. These findings highlight a strong preference for identity verification within the student community.

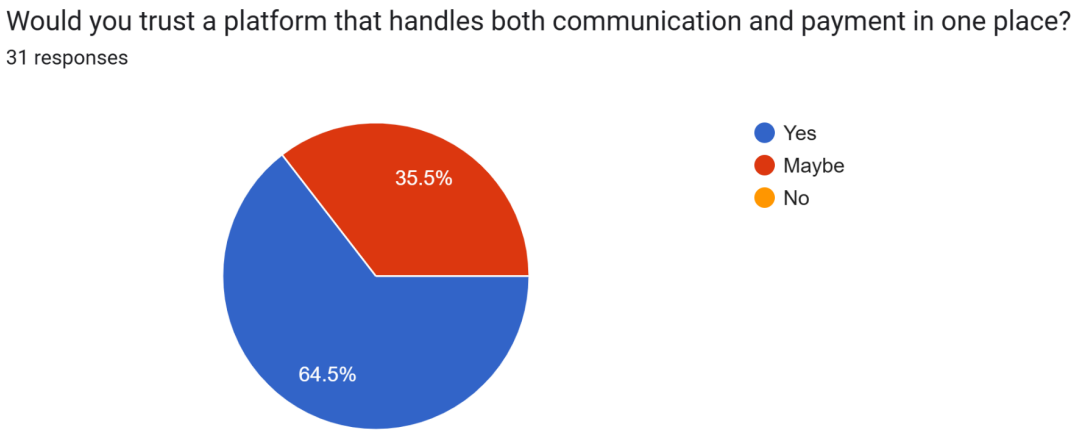


Figure 3.17: Respondents' trust toward an integrated university preloved platform

Figure 3.17 shows that a majority of university students (64.5%) would trust a platform that integrates both communication and payment features while 35.5% responded maybe. No respondents selected no, indicating a generally positive attitude toward unified systems. This suggests strong potential for an all-in-one platform experience where users can interact and transact securely without relying on external apps.

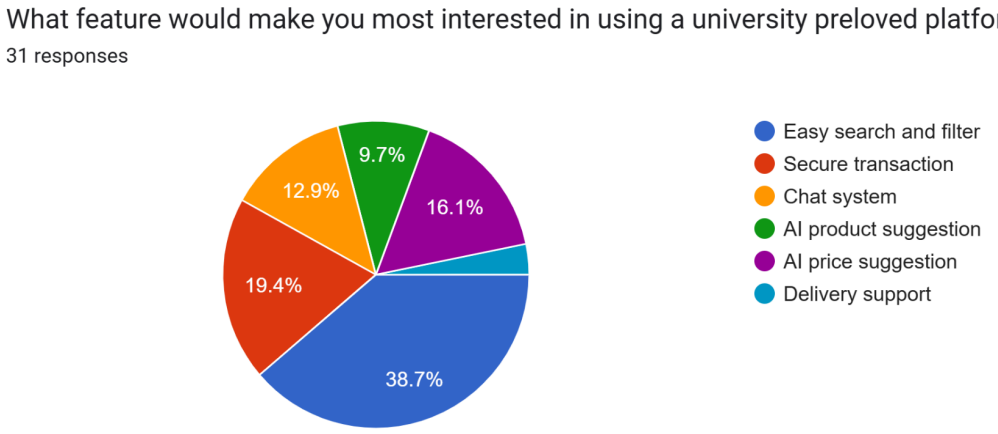


Figure 3.18: Desired features in a university preloved platform

Figure 3.18 shows that the most appealing feature for university students is easy search and filter functionality (38.7%), followed by secure transactions (19.4%) and price suggestion tools (16.1%). Other features such as chat systems (12.9%), AI product suggestions (9.7%) and delivery support (3.2%) were less prioritized. These results indicate that students value efficiency

and safety when navigating preloved platforms. To meet these expectations, the platform should focus on developing robust search capabilities and secure transaction mechanisms while offering optional enhancements like smart recommendations and delivery coordination.

4.0 PROBLEM STATEMENT

The development of ThriftIn UTM is driven by a deep analysis of four core problems currently faced by the student community and provides a technical solution for each.

The issue of trust and identity is one of the most important. Currently, due to the public nature of existing digital groups, non-students can easily participate, leading to scams or unreliable transactions. The solution is the implementation of Mandatory UTM Student Email Authentication, requiring users to register and verify their identity using a genuine university email address (@graduate.utm.my), thereby ensuring the platform remains a secure, closed community exclusively for verified UTM affiliates.

The second problem lies in user experience, specifically how product information is presented and consumed. Sales posts are quickly lost in the noise of busy chat feeds, and there is no capability for structured searching, categorization, or filtering. This is resolved by dedicating a Product Market subsystem with a data structure for categorization, advanced filtering capabilities, and a clear, standardized product listing layout, providing a focused e-commerce experience.

Thirdly, the problem of efficiency and moderation affects group administrators who waste valuable time monitoring off-topic sales posts and often resorting to removing students. ThriftIn UTM addresses this by being a self-contained platform, centralizing all commerce activities into a single, specialized environment, which completely eliminates the burden of external content moderation for existing communication channels.

Finally, the challenge of discovery limits a student's ability to find specific items, such as specific course textbooks or unique lab equipment. To overcome this, the platform will integrate an AI Recommendation Engine that utilizes machine learning principles to suggest relevant products to students based on their recorded course enrollment data, past search history, and items they have added to a personalized wish list, dramatically improving the likelihood of a successful transaction.

5.0 GOAL AND OBJECTIVE

The overall goal of this project is to successfully design, develop and deploy a secure, efficient, and intelligent second-hand trading platform, named ThriftIn UTM that promotes sustainable commerce and enhances affordability exclusively for the benefit of the Universiti Teknologi Malaysia student body. The successful realization of this goal is dependent upon achieving five core technical objectives.

1. Implementation of an authentication and authorization framework centered around the critical UTM email verification process to secure the user base.
2. Creation of the core Product Market module, which must not only support secure listing management but also integrate a functional, secure, in-app chat system to facilitate negotiation.
3. Involves the integration of an AI-powered recommendation system designed to personalize the user experience and significantly improve product discovery by matching buyers with relevant listings.
4. Establish a comprehensive Analytics Dashboard which platform administrators can use to actively monitor user activity, sales performance metrics, and identify high-demand product categories.
5. Ensure the complete application is developed to be fully responsive and adaptable, ready for seamless deployment.

6.0 PROPOSED APPLICATION

Figure 6.1: Logo of the ThriftIn UTM

Connecting all UTM communities within a single platform and simplifying the process of selling or buying used items that are still in good condition, a mobile application, ThriftIn UTM is proposed to be developed. This platform ensures secure transactions and eliminates the issue of being treated as "off-topic", which often occurs on existing informal platforms. ThriftIn UTM not only provides opportunities to strengthen student entrepreneurship but also promotes a sustainable reuse culture within the campus. This innovative solution offers an integrated and enhanced experience for the UTM community to buy and sell preloved items efficiently. The proposed application architecture will be built upon a modular structure, segmented into four interconnected subsystems, ensuring maintainability and clear development responsibilities for the team. Below is the overview of the subsystems and its modules.

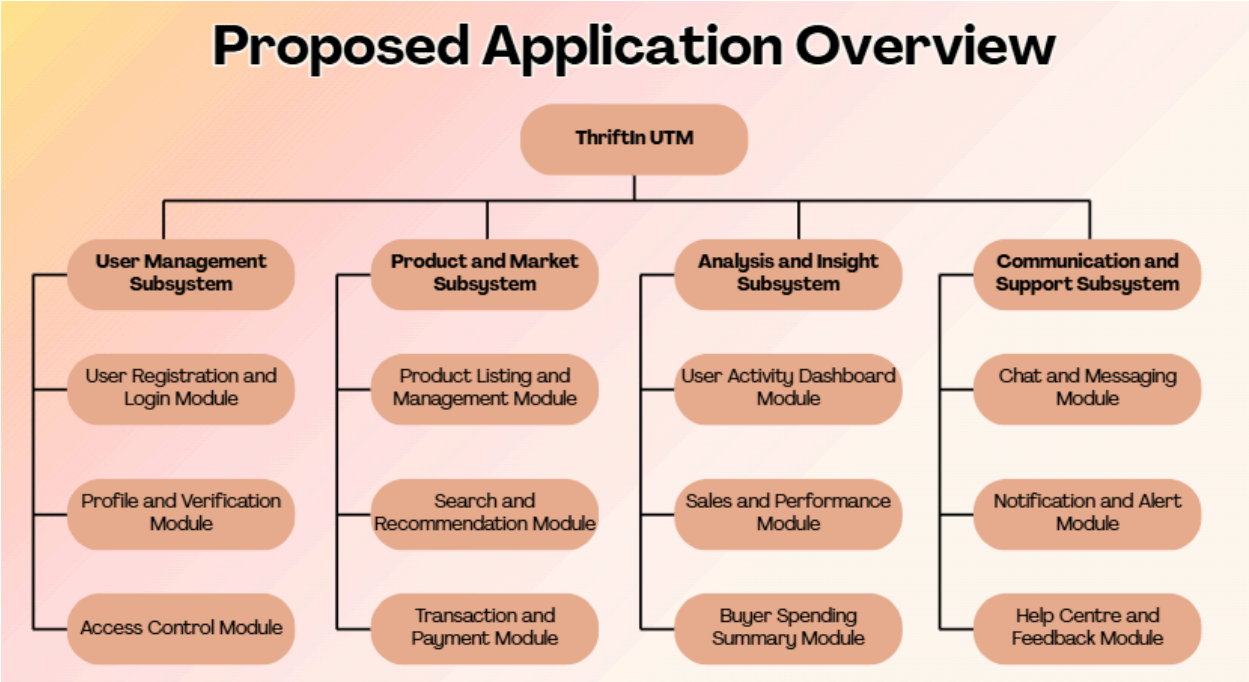


Figure 6.2: Overview of subsystems and modules

The proposed subsystems are designed to enhance communication, improve the transaction process, and connect buyers and sellers within the appropriate target market. The platform integrated key technologies including authorization, authentication, payment gateway, data analytics and AI integration to ensure security, efficiency, and intelligent functionality across the platform.

1. **User Management Subsystem**

   This subsystem manages all user-related activities such as registration, authentication, and profile management. It ensures secure access and proper role-based authorization across the platform. This subsystem provides authentication for all other subsystems and shares verified user data with Product & Market and Communication subsystems.

   Modules:

   - User Registration & Login: Enables new users to sign up and existing users to log in securely.
   - Profile & Verification: Allows users to manage their personal details and verify their identity.
   - Access Control & Authorization: Controls user permissions based on role and verification status, ensuring only authorized users can perform restricted actions.

2. **Product and Market Subsystem**

   This subsystem manages product listings, recommendations, and transactions within the platform, allowing users to browse, search, and buy or sell preloved items easily and intelligently. This subsystem requires authentication from the User Management subsystem and will send listing and sales data to the Analytics subsystem.

   Modules:

   - Product Listing & Management: Sellers can create, edit, and manage product details.
   - Search & Recommendation: Uses filters and AI suggestions based on user preferences or history.
   - Transaction & Payment: Handles payment processing, order confirmation, and transaction history.

3. **Analytics and Insights Subsystem**

   The subsystem analyzes user and product data, providing valuable insights for decision-making and performance tracking. This subsystem collects data from User Management and Product & Market subsystems and will generate reports that can be viewed by admins or sellers.

   Modules:

   - User Activity Dashboard: Displays active users, sessions, and usage patterns.
   - Sales & Performance: Tracks revenue, popular products, and sales trends.
   - Buyer Spending Summary Module: Uses transaction data to summarize and visualize total spending per category.

4. **Communication and Support Subsystem**

   The subsystem provides real-time communication and customer support functionalities, improving user engagement and issue resolution. This subsystem receives user data from User Management and supports feedback analysis that can be sent to the Analytics subsystem.

   Modules:

   - Chat & Messaging: Enables direct communication between users or with support staff.
   - Notification & Alert: Sends real-time updates about transactions, messages, or system alerts.
   - Help Center & Feedback: Allows users to submit feedback, report issues, or read FAQs.

7.0 PROPOSED TECHNOLOGY AND TOOLS USED

Component	Tools	Purpose and Usage
Frontend	 React Native	- Build a cross-platform mobile app (iOS and Android) using JavaScript. - Used to create interactive front-end features such as navigation, forms, buttons, and maps.
	 Figma	- Used to design and prototype the user interface before development. - Simple drag and drop interface for layout planning and team collaboration.
Backend	 Node.js	- A JavaScript runtime environment that executes the backend code. - Allow JavaScript to run outside the application.. - Enable server-side programming and efficient handling of asynchronous operations.
		- A minimalist web framework built on top of Node.js, used as the backend for the mobile app.. - Used to develop the main API and handle core business logic

	Express.js	such as user authentication (UTM email verification), session management, product CRUD operations (Create, Read, Update, Delete) and connecting to the MySQL database.
Database	 MySQL	- An open-source relational database used to store and manage all platform data, including user profiles, product listings and transactions. - Enable seamless interaction between subsystems.
API Communications	 Axios	- A promise-based HTTP client used by the React frontend to make API calls to the Node.js and FastAPI backends. - Handle sending data and receiving responses efficiently and reliably.
AI Product Recommendation and Price Suggestion	 FastAPI	- A modern, fast Python web framework used to create a separate microservice for AI/ML tasks in the mobile app backend. - Provide dedicated endpoints for product recommendation models and price suggestion algorithms,

		ensuring these computationally intensive tasks do not slow down the main Node.js API.
Project Management and Collaboration	Trello	- Organize sprints, tasks and progress with boards and cards. - Encourage teamwork and transparent communication.
	GitHub	- Manage the project's code repository online. - Enable version control, collaboration and automatic backups. - Simplify commits and pushes.
Testing	Expo	- Let us run and test mobile app on Android and iOS without installing native SDKs. - Allow us to see UI changes instantly, interact with forms, buttons and backend APIs.

Table 7.1: Proposed technology and tools used

8.0 PROJECT BENEFITS

The successful deployment of ThriftIn UTM will deliver substantial benefits across various stakeholders.

For the UTM Students, the platform provides access to a fully secure and verified marketplace, which inherently leads to higher transaction reliability, increased trust among buyers and sellers, and a significantly easier process for discovering specific items like academic materials or lab equipment. It also allows students to buy second-hand items at affordable prices. This helps reduce financial pressure, especially for new or low-income students and promotes a budget-friendly shopping option within campus. Additionally, it encourages students to reuse and recycle items instead of throwing them away.

Kelab Keusahawanan UTM gains a professionally developed, dedicated digital platform to manage, monitor, and facilitate all student commerce activities. This directly advances the club's mission to strengthen student entrepreneurship by providing a specialized, data-rich tool for overseeing trade and generating insights into student market behavior.

The UTM Administration benefits by having a dedicated, self-managed commerce channel, which effectively reduces the clutter and volume of off-topic posts in official communication channels, resulting in cleaner, more efficient administrative messaging and an improved overall digital community experience.

As a result, the platform makes contributions to sustainability by actively promoting a circular economy on campus, thereby extending the usable lifecycle of various items and aligning the university community with broader environmental responsibility goals.

9.0 PROJECT SCHEDULE

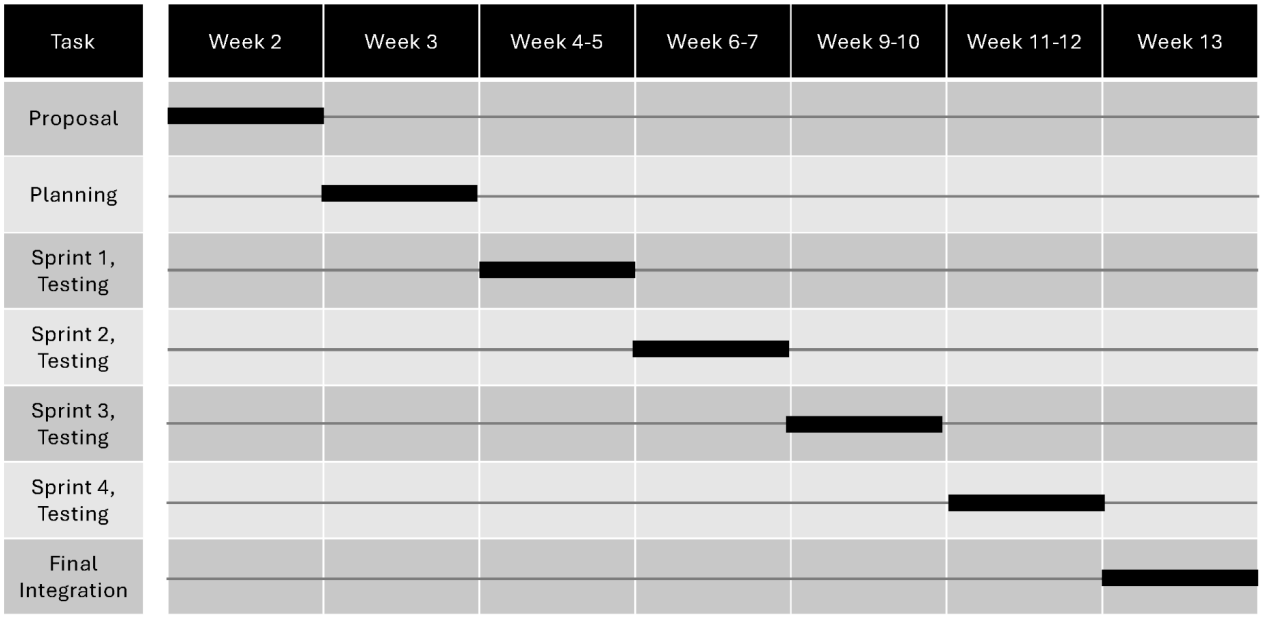


***Figure 10.1*: Gantt Chart for Overall Project Development**

10.0 TEAM MEMBERS AND ROLES

10.1 SUBSYSTEMS AND RESPONSIBLE TEAM MEMBERS

Subsystem	Person In Charge
User Management Subsystem	Angie
Product and Market Subsystem	Aina
Analytics and Insights Subsystem	Asilah
Communication and Support Subsystem	Jia Ying

Table 10.1*: *Distribution of Subsystems

10.2 SCRUM ROLES

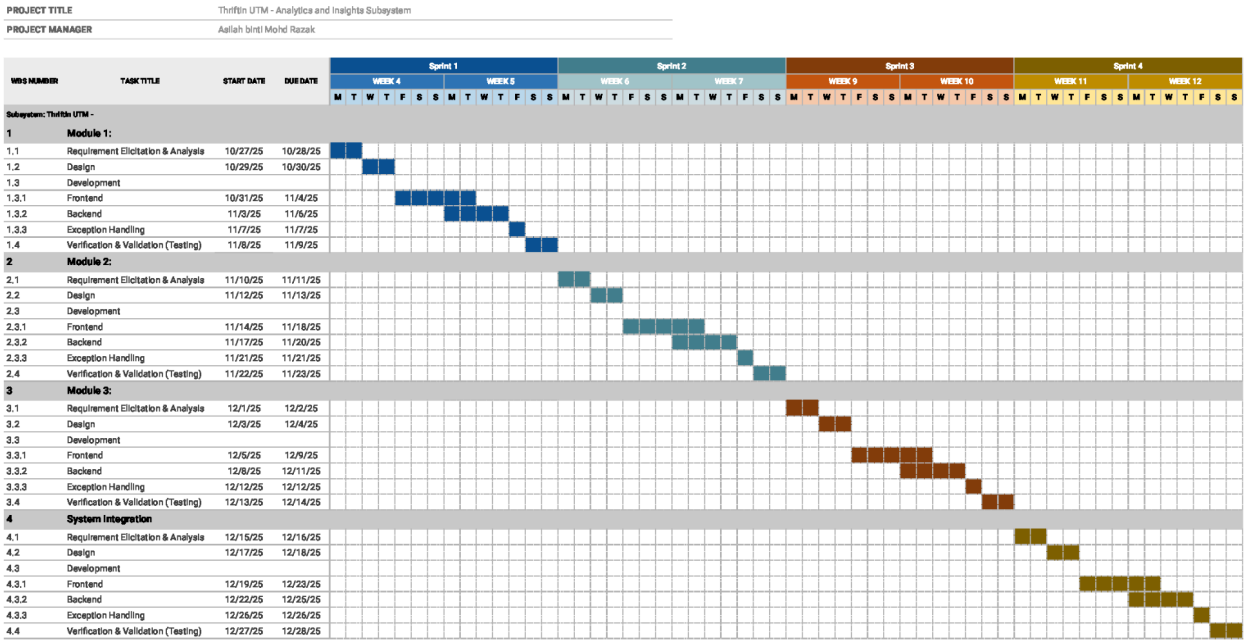
	Sprint 1	Sprint 2	Sprint 3	Sprint 4
Project Manager	Asilah	Angie	Aina	Jia Ying
Scrum Master	Angie	Aina	Jia Ying	Asilah
Developer	Aina	Jia Ying	Asilah	Angie
Tester	Jia Ying	Asilah	Angie	Aina

Table 10.2: Roles of Each Member for Each Sprint

11.0 INDIVIDUAL SCHEDULE

11.1 ASILAH BINTI MOHD RAZAK

PDF link: AD Individual Schedule - ASILAH MOHD RAZAK.pdf



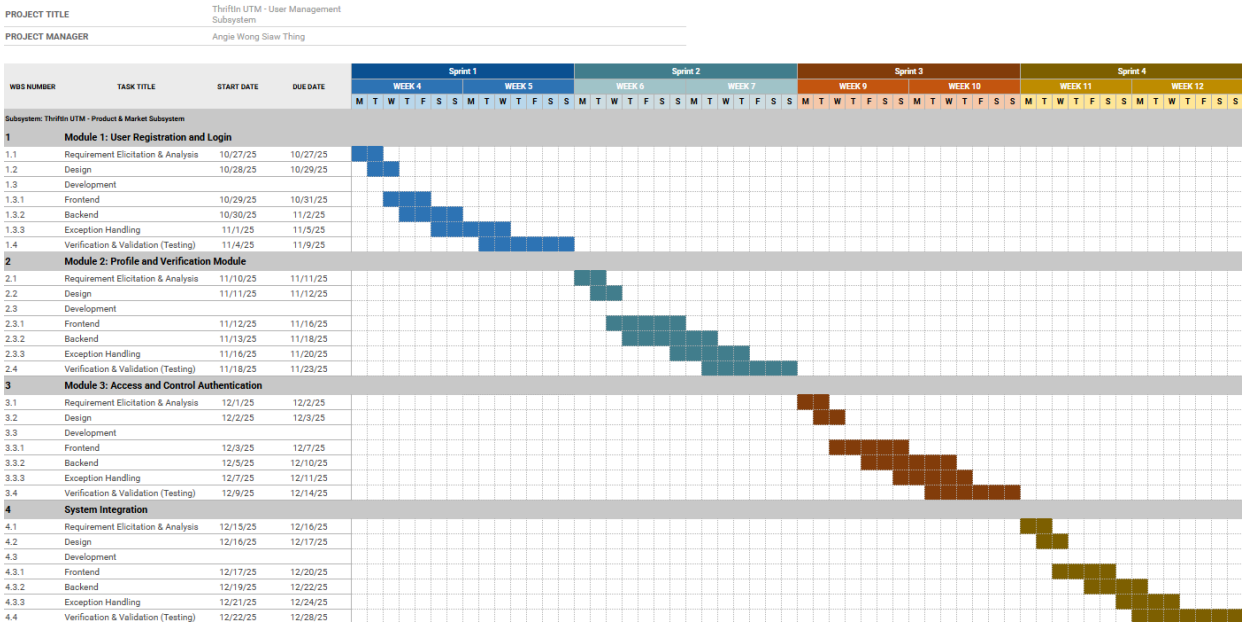
PROJECT TITLE	Thriftin UTM - Analytics and Insights Subsystem
PROJECT MANAGER	Asilah binti Mohd Razak

WBS NUMBER	TASK TITLE	START DATE	DUE DATE
Subsystem: Thriftin UTM -			
1	**Module 1:**		
1.1	Requirement Elicitation & Analysis	10/27/25	10/28/25
1.2	Design	10/29/25	10/30/25
1.3	Development		
1.3.1	Frontend	10/31/25	11/4/25
1.3.2	Backend	11/3/25	11/6/25
1.3.3	Exception Handling	11/7/25	11/7/25
1.4	Verification & Validation (Testing)	11/8/25	11/9/25
2	**Module 2:**		
2.1	Requirement Elicitation & Analysis	11/10/25	11/11/25
2.2	Design	11/12/25	11/13/25
2.3	Development		
2.3.1	Frontend	11/14/25	11/18/25
2.3.2	Backend	11/17/25	11/20/25
2.3.3	Exception Handling	11/21/25	11/21/25
2.4	Verification & Validation (Testing)	11/22/25	11/23/25
3	**Module 3:**		
3.1	Requirement Elicitation & Analysis	12/1/25	12/2/25
3.2	Design	12/3/25	12/4/25
3.3	Development		
3.3.1	Frontend	12/5/25	12/9/25
3.3.2	Backend	12/8/25	12/11/25
3.3.3	Exception Handling	12/12/25	12/12/25
3.4	Verification & Validation (Testing)	12/13/25	12/14/25
4	**System Integration**		
4.1	Requirement Elicitation & Analysis	12/15/25	12/16/25
4.2	Design	12/17/25	12/18/25
4.3	Development		
4.3.1	Frontend	12/19/25	12/23/25
4.3.2	Backend	12/22/25	12/25/25
4.3.3	Exception Handling	12/26/25	12/26/25
4.4	Verification & Validation (Testing)	12/27/25	12/28/25

Figure 11.1: The Individual Schedule of Asilah Binti Mohd Razak

11.2 ANGIE WONG SIAW THING

PDF link: Individual Schedule - Angie Wong Siaw Thing.pdf



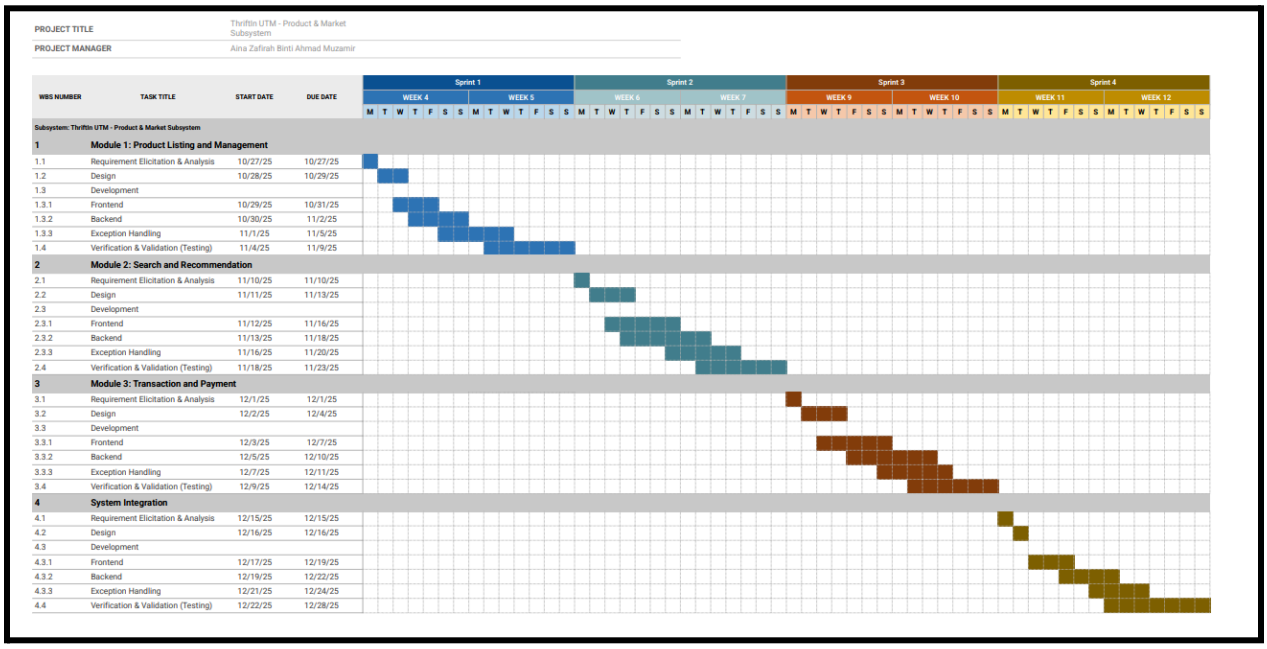
PROJECT TITLE	ThriftIn UTM - User Management Subsystem
PROJECT MANAGER	Angie Wong Siaw Thing

WBS NUMBER	TASK TITLE	START DATE	DUE DATE
Subsystem: ThriftIn UTM - Product & Market Subsystem			
1	**Module 1: User Registration and Login**		
1.1	Requirement Elicitation & Analysis	10/27/25	10/27/25
1.2	Design	10/28/25	10/29/25
1.3	Development		
1.3.1	Frontend	10/29/25	10/31/25
1.3.2	Backend	10/30/25	11/2/25
1.3.3	Exception Handling	11/1/25	11/5/25
1.4	Verification & Validation (Testing)	11/4/25	11/9/25
2	**Module 2: Profile and Verification Module**		
2.1	Requirement Elicitation & Analysis	11/10/25	11/11/25
2.2	Design	11/11/25	11/12/25
2.3	Development		
2.3.1	Frontend	11/12/25	11/16/25
2.3.2	Backend	11/13/25	11/18/25
2.3.3	Exception Handling	11/16/25	11/20/25
2.4	Verification & Validation (Testing)	11/18/25	11/23/25
3	**Module 3: Access and Control Authentication**		
3.1	Requirement Elicitation & Analysis	12/1/25	12/2/25
3.2	Design	12/2/25	12/3/25
3.3	Development		
3.3.1	Frontend	12/3/25	12/7/25
3.3.2	Backend	12/5/25	12/10/25
3.3.3	Exception Handling	12/7/25	12/11/25
3.4	Verification & Validation (Testing)	12/9/25	12/14/25
4	**System Integration**		
4.1	Requirement Elicitation & Analysis	12/15/25	12/16/25
4.2	Design	12/16/25	12/17/25
4.3	Development		
4.3.1	Frontend	12/17/25	12/20/25
4.3.2	Backend	12/19/25	12/22/25
4.3.3	Exception Handling	12/21/25	12/24/25
4.4	Verification & Validation (Testing)	12/22/25	12/28/25

Figure 11.2: The Individual Schedule of Angie Wong Siaw Thing

11.3 AINA ZAFIRAH BINTI AHMAD MUZAMIR

PDF Link: Individual Schedule - AINA ZAFIRAH.pdf



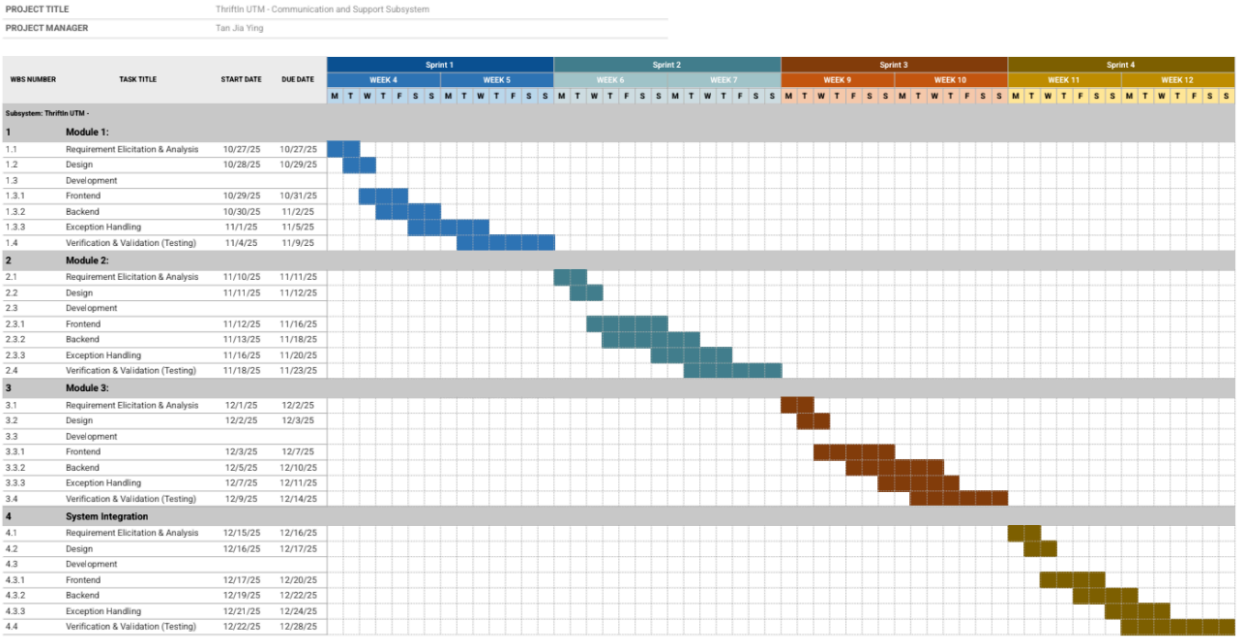
PROJECT TITLE	Thriftin UTM - Product & Market Subsystem
PROJECT MANAGER	Aina Zafirah Binti Ahmad Muzamir

WBS NUMBER	TASK TITLE	START DATE	DUE DATE
Subsystem: Thriftin UTM - Product & Market Subsystem			
1	**Module 1: Product Listing and Management**		
1.1	Requirement Elicitation & Analysis	10/27/25	10/27/25
1.2	Design	10/28/25	10/29/25
1.3	Development		
1.3.1	Frontend	10/29/25	10/31/25
1.3.2	Backend	10/30/25	11/2/25
1.3.3	Exception Handling	11/1/25	11/5/25
1.4	Verification & Validation (Testing)	11/4/25	11/9/25
2	**Module 2: Search and Recommendation**		
2.1	Requirement Elicitation & Analysis	11/10/25	11/10/25
2.2	Design	11/11/25	11/13/25
2.3	Development		
2.3.1	Frontend	11/12/25	11/16/25
2.3.2	Backend	11/13/25	11/18/25
2.3.3	Exception Handling	11/16/25	11/20/25
2.4	Verification & Validation (Testing)	11/18/25	11/23/25
3	**Module 3: Transaction and Payment**		
3.1	Requirement Elicitation & Analysis	12/1/25	12/1/25
3.2	Design	12/2/25	12/4/25
3.3	Development		
3.3.1	Frontend	12/3/25	12/7/25
3.3.2	Backend	12/5/25	12/10/25
3.3.3	Exception Handling	12/7/25	12/11/25
3.4	Verification & Validation (Testing)	12/9/25	12/14/25
4	**System Integration**		
4.1	Requirement Elicitation & Analysis	12/15/25	12/15/25
4.2	Design	12/16/25	12/16/25
4.3	Development		
4.3.1	Frontend	12/17/25	12/19/25
4.3.2	Backend	12/19/25	12/22/25
4.3.3	Exception Handling	12/21/25	12/24/25
4.4	Verification & Validation (Testing)	12/22/25	12/28/25

Figure 11.3: The Individual Schedule of Aina Zafirah Binti Ahmad Muzamir

11.4 TAN JIA YING

PDF Link: Individual Schedule - Jiaying - Individual Schedule.pdf



PROJECT TITLE	Thriftin UTM - Communication and Support Subsystem
PROJECT MANAGER	Tan Jia Ying

WBS NUMBER	TASK TITLE	START DATE	DUE DATE
Subsystem: Thriftin UTM -			
1	**Module 1:**		
1.1	Requirement Elicitation & Analysis	10/27/25	10/27/25
1.2	Design	10/28/25	10/29/25
1.3	Development		
1.3.1	Frontend	10/29/25	10/31/25
1.3.2	Backend	10/30/25	11/2/25
1.3.3	Exception Handling	11/1/25	11/5/25
1.4	Verification & Validation (Testing)	11/4/25	11/9/25
2	**Module 2:**		
2.1	Requirement Elicitation & Analysis	11/10/25	11/11/25
2.2	Design	11/11/25	11/12/25
2.3	Development		
2.3.1	Frontend	11/12/25	11/16/25
2.3.2	Backend	11/13/25	11/18/25
2.3.3	Exception Handling	11/16/25	11/20/25
2.4	Verification & Validation (Testing)	11/18/25	11/23/25
3	**Module 3:**		
3.1	Requirement Elicitation & Analysis	12/1/25	12/2/25
3.2	Design	12/2/25	12/3/25
3.3	Development		
3.3.1	Frontend	12/3/25	12/7/25
3.3.2	Backend	12/5/25	12/10/25
3.3.3	Exception Handling	12/7/25	12/11/25
3.4	Verification & Validation (Testing)	12/9/25	12/14/25
4	**System Integration**		
4.1	Requirement Elicitation & Analysis	12/15/25	12/16/25
4.2	Design	12/16/25	12/17/25
4.3	Development		
4.3.1	Frontend	12/17/25	12/20/25
4.3.2	Backend	12/19/25	12/22/25
4.3.3	Exception Handling	12/21/25	12/24/25
4.4	Verification & Validation (Testing)	12/22/25	12/28/25

Figure 11.4: The Individual Schedule of Tan Jia Ying